Bhagyalekshmy S

Research Intern
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Homepage

Education

- **BS-MS Dual Degree** *2020 - 2025*
  Indian Institute of Science Education and Research, Pune
  - **MS in Mathematics**
    Thesis: Topics in Code-based Cryptography
    CGPA: 9/10
  - **BS in Mathematics**
    CGPA: 7.1/10

Research Experience

• Internship *July 2026 - present*

Supervised by Dr Rajendra Kumar, IIT Delhi

Lattice-based cryptography: This project involved a comprehensive study of the learning with errors (LWE) problem, extrapolated dihedral coset problems (EDCP), and their algebraic variants. It also included analyzing the quantum reductions between Module-LWE and a structured variant of EDCP (IP-M-EDCP).

• Research Project *November 2025 - present*

Project with Dr Rutuja Kshirsagar, Fujitsu Research of America Inc., Prof Henry Chimal-Dzul, UTSA and Angelo Lomeli, UTSA

Code-based Cryptography: This work focuses on quasi-twisted (QT) code constructions. The following are the two main directions explored in this project.

- Viewing quasi-twisted codes from the perspectives of the extension fields. We generalize previous results on quasi-cyclic (QC) codes in extension fields under certain conditions and eliminate the said conditions so as to include all QC and QT codes.
- Addressing the computational aspects of my master's thesis.

• Master's Thesis Research Project (year-long project) *May 2024 - May 2025*

Supervised by Dr Krishna Kaipa, IISER Pune and Dr Rutuja Kshirsagar, Fujitsu Research of America Inc. ***GPA: 9/10***

Topics in Code-based Cryptography: In this thesis, we analyzed the research problem of replacing the traditionally used Goppa codes in the McEliece/Niederreiter cryptosystem with a suitable linear code, aiming to obtain a quantum-secure code-based cryptosystem with a reduced key size. The following are the key results obtained in the course of this project:

- Formalized a bound on the minimum distance of quasi-twisted codes.
- Proposed a syndrome-based decoding algorithm for quasi-twisted codes.
- Developed a Niederreiter-like cryptosystem based on quasi-twisted codes.

We prove that our cryptosystem is secure against QFS-based quantum attacks.

• Theory Residency Program *December 2024*

Project with Dr Rajendra Kumar, CS Theory, IIT Delhi

Fine-grained Cryptography: This program focused on understanding the foundations of fine-grained cryptography and delving into state-of-the-art research problems in the field. The latter was accomplished by studying a recent publication titled, "$k-$SUM in the Sparse Regime".

• Semester Project *Fall semester 2023*

Project with Dr Kaneenika Sinha, IISER Pune

Number Theory: In this project, I studied the basics of number theory, keeping "Introduction to Number Theory" by L.K. Hua as the main reference.

• **Summer Internship** *Summer 2023*

Project at Arqanum Technologies, Pune
This project focused on the following areas:

* Reviewed literature in the field of symmetric cryptography, quantum cryptography and blockchain.
* Analysed block ciphers such as the PRINCE cipher and the AES cipher.
* Implemented the S-boxes of light-weight ciphers on Qiskit (Python).
* Attempted to implement the AES cipher on Qiskit. (GitHub link)

A part of this project also involved understanding codes written in C++.

Publications

1. Bhagyalekshmy S and Rutuja Kshirsagar, "Quasi-twisted codes and their applications in code-based cryptography." (https://arxiv.org/pdf/2507.01118)
2. Prof Henry Chimal-Dzul, Dr Rutuja Kshirsagar, Angelo Lomeli and Bhagyalekshmy S, "An Extension Field Framework for Quasi-Twisted Codes and Matrix-Product Codes." *(under submission)*
3. Prof Henry Chimal-Dzul, Dr Rutuja Kshirsagar, Angelo Lomeli and Bhagyalekshmy S, computational works related to my thesis work on quasi-twisted codes (*work in progress*).

Workshops

Games at Mumbai 2024, Combinatorial workshop, IEOR, IIT Bombay *January 2024*

IIT Delhi TCS Winter School, CS Theory, IIT Delhi *December 2023*

IBM Qiskit Fall Fest'23, DIAT, Pune *October 2023*

TIFR SWIM 2022, TIFR Centre For Applicable Mathematics, Bengaluru *June 2022*

Technical Skills

Programming: Python, SageMath, HTML, SQL

Tools/Frameworks: Qiskit, Keras, Tensorflow

Others: MS Excel, Word, Powerpoint

Non-technical Skills

Soft skills: Writing, Problem solving & Analytic skills

Languages: English, Hindi, Malayalam

Positions of Responsibility

- **Coordinator:** Design Head, Kalpa, IISER Pune Dec. 2022 - Nov. 2023
- **Volunteer:** Social Initiatives & Logistics, Karavaan, IISER Pune Aug. 2022 - Nov. 2022
- **Volunteer:** Design Team, Kalpa, IISER Pune 2021

Academic Achievements

- INSPIRE-SHE Scholarship, Scholarship for Higher Education (SHE) *2020, 2024*
- 100 percent marks in class 12 Board Exams for Mathematics *2020*
- CBSE 0.1 Merit Certificate for Mathematics *2020*
- CBSE 0.1 Merit Certificate for English Communicative *2018*

References

1. **Prof Henry Chimal-Dzul**
   Assistant Professor
   Department of Mathematics, University of Texas at San Antonio, USA
   Research Project Supervisor
   Henry.Chimal-Dzul@utsa.edu

2. **Dr Rutuja Kshirsagar**
   Researcher
   Fujitsu Research of America, Inc., Santa Clara, California
   Master's Thesis Supervisor
   rkshirsagar@fujitsu.com

3. **Dr Krishna Kaipa**
   Associate Professor
   Department of Mathematics, IISER Pune, India
   Master's Thesis Supervisor
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